1. Boston House Price Prediction Dataset

The **Boston House Price Prediction Dataset** is a classic and widely used dataset in **machine learning and data science**, especially for learning **regression techniques**. It was originally collected by the **U.S. Census Service** and contains information about housing values in suburbs of **Boston, Massachusetts**.

Purpose

The dataset is mainly used to **predict the median value of owner-occupied homes** based on various socio-economic and environmental factors. It helps beginners and researchers understand **linear regression, multiple regression, feature importance, and model evaluation**.

Dataset Size

- **Total records:** 506 houses
- **Number of features:** 13 input features
- **Target variable:** 1 (house price)

Features (Independent Variables)

1. **CRIM** – Per capita crime rate by town
2. **ZN** – Proportion of residential land zoned for large plots
3. **INDUS** – Proportion of non-retail business acres
4. **CHAS** – Charles River dummy variable (1 if tract bounds river, else 0)
5. **NOX** – Nitric oxide concentration (air pollution)
6. **RM** – Average number of rooms per dwelling
7. **AGE** – Proportion of owner-occupied units built before 1940
8. **DIS** – Distance to employment centers
9. **RAD** – Accessibility to radial highways
10. **TAX** – Property tax rate
11. **PTRATIO** – Pupil–teacher ratio
12. **B** – Proportion of Black population
13. **LSTAT** – Percentage of lower-status population

Target Variable (Dependent Variable)

- **MEDV** – Median value of owner-occupied homes (in $1000s)

Key Characteristics

- Numerical dataset with **no categorical variables**
- Suitable for **supervised learning**
- Used for **regression**, not classification
- Requires **feature scaling** for some algorithms
- Contains **outliers** and correlated features

Common Algorithms Used

- Linear Regression
- Multiple Linear Regression
- Ridge & Lasso Regression
- Decision Tree Regression
- Random Forest Regression
- Support Vector Regression (SVR)

Applications

- Academic learning and practice
- Regression model comparison
- Feature importance analysis
- Exploratory data analysis (EDA)

Limitations

- Small dataset (not suitable for deep learning)
- Contains **ethical concerns** related to feature **B**
- No longer recommended for real-world policy decisions